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Internship Domain: Data Science
Task Number: 03

Technologies Used:

Python, Pandas, SQL, Power BI and google colab

Dataset Description:

It consists of exactly 51290 records and 24 features.

Dataset Source:

<https://www.kaggle.com/datasets/jr2ngb/superstore-data>

Based on an end-to-end data audit of **Sample - Superstore Cleaned.csv** (20,067 records across 13 regions), here is a strategic breakdown of 15 business insights and 10 actionable business recommendations.

15 Core Business Insights

Sales & Revenue Generation

1. **Strong Global Sales Volume:** The business generated **\$4,865,522.83** in total sales with an overall profit margin of **11.84%** (\$576,114.76 net profit).
2. **Category Dominance: Technology** leads all product divisions in gross revenue at **\$1,869,539.35**, followed by **Furniture** (\$1,537,276.17) and **Office Supplies** (\$1,458,707.31).
3. **Core Profit Engine: Copiers** and **Phones** are the primary revenue drivers, generating **\$599,709.73** and **\$650,587.45** in total sales respectively.
4. **Volume vs. Profit Disconnect:** High product volume in **Office Supplies** (68% of order transactions) yields low unit ticket sizes, generating high steady revenue but modest overall dollar profits per transaction compared to Technology.
5. **Customer Concentration Balance:** Revenue is well-distributed across the customer base, with top spender **Tamara Chand** accounting for **\$25,679.70** (less than 0.6% of overall sales).

Profitability & Pricing Strategy

6. **Severe Table Line Erosion:** **Tables** is the single worst-performing sub-category, generating a **-\$22,425.75 loss** on \$281,497.76 in total sales due to an aggressive average discount rate of **28.63%**.
7. **Zero-Discount Margin Strength:** Transactions with **No Discount** deliver **\$698,719.36** in net profit at a peak margin of **25.68%**.

8. **High Discount Destruction:** Transactions in the **High Discount** band result in an extreme negative margin of **-114.49%**, burning **\$157,779.35** in gross capital.
9. **Heavy Profit Drag:** Across the entire catalog, 4,913 individual orders resulted in negative profitability, draining **\$355,950.02** from potential net earnings.
10. **High-Margin Staples: Paper** (23.99% margin) and **Labels** (21.70% margin) deliver consistent, low-risk profits with minimal return or discount erosion.

Regional & Operational Performance

11. **Regional Anchor: Central** is the largest geographical revenue contributor (**\$1,053,612.44** in total sales), maintaining a stable **11.23%** profit margin.
12. **High-Efficiency Markets: North Asia** and **Central Asia** represent the highest-margin sales territories, operating at **20.17%** and **19.27%** net profit margins respectively.
13. **Southeast Asian Margin Compression: Southeast Asia** generates **\$360,149.60** in sales but achieves an unviable net profit margin of only **2.31%** due to local discounting and freight overheads.
14. **EMEA Underperformance:** The **EMEA** region suffers from depressed margins (**6.63%**) despite running 995 distinct orders, indicating high fulfillment costs relative to order sizes.
15. **Premium Freight Monetization: First Class** and **Same Day** shipping modes maintain healthy contribution margins, demonstrating strong customer willingness to pay full rate for express fulfillment.

10 Strategic Business Recommendations

Pricing & Discount Governance

1. **Implement Hard Discount Caps:** Immediately enforce a firm **15% ceiling** on order-level discounts across all categories. Disallow any discounts above 20% without regional VP approval.
2. **Restructure Tables Pricing Strategy:** Restructure or drop low-margin items within the **Tables** sub-category. Eliminate baseline discounting on Tables to reverse its current **-\$22,425.75** net loss.
3. **Protect Non-Discounted Lines:** Maintain baseline pricing across **Paper**, **Labels**, and **Copiers**, using non-price incentives (such as bundled express freight) rather than cash discounts to close corporate deals.

Geographic Optimization & Regional Strategy

4. **Overhaul Southeast Asian Logistics:** Audit logistics providers, landed import tariffs, and local distribution channels in **Southeast Asia** to lift profit margins from **2.31%** toward the global benchmark of **11.84%**.

5. **Expand in High-Margin Markets:** Increase marketing spending and account executive coverage in **North Asia** (20.17% margin) and **Central Asia** (19.27% margin), where gross margins are highest.
6. **Rationalize EMEA Operations:** Set minimum order thresholds (MOQs) for **EMEA** orders to consolidate freight charges and eliminate thin-margin individual shipments.

Catalog & Sales Portfolio Management

7. **Cross-Sell High-Margin Add-ons:** Bundle low-margin items (e.g., Supplies or Furniture) with high-margin items like **Copiers** (17.80% margin) and **Accessories** (16.80% margin) to preserve basket profitability.
8. **Automate Reorders on Fast-Moving Office Supplies:** Convert high-volume **Paper** and **Storage** buyers into automated subscription contracts to guarantee predictable recurring income.

Customer & Channel Strategy

9. **VIP Retention Program for Top Accounts:** Establish key account management protocols for top-tier corporate buyers (e.g., *Tamara Chand*, *Bart Watters*) offering tier-based volume rebates instead of upfront line-item discounts.
10. **Eliminated Negative-Margin Promotional Triggers:** Adjust automated ecommerce checkout algorithms to disable stacking discounts on already-discounted items or orders requiring expensive air shipping.

GitHub Repository Link:

<https://github.com/abdulwahab014-CS/Superstore-Growth-Tracker>